



中山大學
SUN YAT-SEN UNIVERSITY

数学分析

第十六章 偏微分与全微分 §2 复合函数与隐函数微分法

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链式规则

链式规则

设 $z = f(x, y)$, $(x, y) \in \mathbf{D}_f$ 是区域 $\mathbf{D}_f \subset \mathbb{R}^2$ 上的二元函数, 而

$$\mathbf{g}: \mathbf{D}_g \rightarrow \mathbb{R}^2, \quad (u, v) \mapsto (x(u, v), y(u, v))$$

是区域 $\mathbf{D}_g \subset \mathbb{R}^2$ 上的二元二维向量值函数。如果 $\mathbf{g}$ 的值域 $\mathbf{g}(\mathbf{D}_g) \subset \mathbf{D}_f$, 那么可以构造复合函数

$$z = f \circ \mathbf{g} = f[x(u, v), y(u, v)], \quad (u, v) \in \mathbf{D}_g.$$

复合函数有如下求偏导数的法则。

链式规则（续）

链式规则（续）

设 $\mathbf{g}$ 在 $(u_0, v_0) \in \mathbf{D}_g$ 点可导, 即 $x = x(u, v)$, $y = y(u, v)$ 在 (u_0, v_0) 点可偏导。记 $x_0 = x(u_0, v_0)$, $y_0 = y(u_0, v_0)$, 如果 f 在 (x_0, y_0) 点可微, 那么

$$\frac{\partial z}{\partial u}(u_0, v_0) = \frac{\partial z}{\partial x}(x_0, y_0) \frac{\partial x}{\partial u}(u_0, v_0) + \frac{\partial z}{\partial y}(x_0, y_0) \frac{\partial y}{\partial u}(u_0, v_0);$$

$$\frac{\partial z}{\partial v}(u_0, v_0) = \frac{\partial z}{\partial x}(x_0, y_0) \frac{\partial x}{\partial v}(u_0, v_0) + \frac{\partial z}{\partial y}(x_0, y_0) \frac{\partial y}{\partial v}(u_0, v_0).$$

链式规则的证明

证 只证明第一式。由于 f 在 (x_0, y_0) 点可微, 因此

$$\begin{aligned} & f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0) \\ &= \frac{\partial f}{\partial x}(x_0, y_0)\Delta x + \frac{\partial f}{\partial y}(x_0, y_0)\Delta y + \underbrace{\frac{o(\sqrt{\Delta x^2 + \Delta y^2})}{\sqrt{\Delta x^2 + \Delta y^2}}}_{\alpha(\Delta x, \Delta y)} \sqrt{\Delta x^2 + \Delta y^2}, \end{aligned}$$

其中 $\alpha(\Delta x, \Delta y)$ 满足 $\lim_{(\Delta x, \Delta y) \rightarrow 0} \alpha(\Delta x, \Delta y) = 0$ 。定义 $\alpha(0, 0) = 0$, 那么上式当 $(\Delta x, \Delta y) = (0, 0)$ 时也成立。

链式规则的证明 (续)

设

$$\Delta x = x(u_0 + \Delta u, v_0) - x(u_0, v_0), \quad \Delta y = y(u_0 + \Delta u, v_0) - y(u_0, v_0),$$

由于 $x = x(u, v)$, $y = y(u, v)$ 在 (u_0, v_0) 点可偏导, 所以成立

$$\lim_{\Delta u \rightarrow 0} \frac{\Delta x}{\Delta u} = \frac{\partial x}{\partial u}(u_0, v_0), \quad \lim_{\Delta u \rightarrow 0} \frac{\Delta y}{\Delta u} = \frac{\partial y}{\partial u}(u_0, v_0),$$

并且有 $\lim_{\Delta u \rightarrow 0} \sqrt{\Delta x^2 + \Delta y^2} = 0$ 。于是当 $\Delta u \rightarrow 0$ 时,

$$\frac{\alpha(\Delta x, \Delta y) \sqrt{\Delta x^2 + \Delta y^2}}{\Delta u} = \alpha(\Delta x, \Delta y) \cdot \frac{|\Delta u|}{\Delta u} \cdot \sqrt{\left(\frac{\Delta x}{\Delta u}\right)^2 + \left(\frac{\Delta y}{\Delta u}\right)^2}$$

也趋于 0, 所以

链式规则的证明 (续)

$$\begin{aligned}\frac{\partial z}{\partial u}(u_0, v_0) &= \lim_{\Delta u \rightarrow 0} \frac{f(x(u_0 + \Delta u, v_0), y(u_0 + \Delta u, v_0)) - f(x(u_0, v_0), y(u_0, v_0))}{\Delta u} \\&= \lim_{\Delta u \rightarrow 0} \frac{f(x_0 + \Delta x, y_0 + \Delta y) - f(x_0, y_0)}{\Delta u} \\&= \lim_{\Delta u \rightarrow 0} \left[\frac{\partial f}{\partial x}(x_0, y_0) \frac{\Delta x}{\Delta u} + \frac{\partial f}{\partial y}(x_0, y_0) \frac{\Delta y}{\Delta u} \right] + \lim_{\Delta u \rightarrow 0} \frac{\alpha(\Delta x, \Delta y) \sqrt{\Delta x^2 + \Delta y^2}}{\Delta u} \\&= \frac{\partial f}{\partial x}(x_0, y_0) \frac{\partial x}{\partial u}(u_0, v_0) + \frac{\partial f}{\partial y}(x_0, y_0) \frac{\partial y}{\partial u}(u_0, v_0).\end{aligned}$$

证毕

链式规则的适用性与反例

注意，定理条件“ f 可微”不能减弱为“ f 可偏导”。

链式规则的适用性与反例

例 1.1 从上节已知,

$$z = f(x, y) = \begin{cases} \frac{2xy^3}{x^2 + y^4}, & x^2 + y^2 \neq 0, \\ 0, & x^2 + y^2 = 0 \end{cases}$$

在 $(0, 0)$ 点可偏导, 且 $f_x(0, 0) = f_y(0, 0) = 0$, 但它在 $(0, 0)$ 点不可微. 设 $x = t^2$, $y = t$, 则

$$z = f(x, y) = f(t^2, t) = t, \quad \text{所以在 } t = 0 \text{ 点的导数 } \frac{dz}{dt}(0) = 1.$$

链式规则的适用性与反例

但若贸然套用链式法则，则得到如下错误结果

$$\begin{aligned}\frac{dz}{dt}(0) &= [f_x(t^2, t) \cdot 2t + f_y(t^2, t) \cdot 1] \Big|_{t=0} \\ &= f_x(0, 0) \cdot 0 + f_y(0, 0) \cdot 1 = 0.\end{aligned}$$

推广到一般情形

下面不加证明地把链式规则推广至一般情况。

设

$$z = f(y_1, y_2, \cdots, y_m), \quad (y_1, y_2, \cdots, y_m) \in \mathbf{D}_f$$

为区域 $\mathbf{D}_f \subset \mathbb{R}^m$ 上的 m 元函数。又设

$$\mathbf{g}: \mathbf{D}_g \rightarrow \mathbb{R}^m, \quad (x_1, x_2, \cdots, x_n) \mapsto (y_1, y_2, \cdots, y_m)$$

为区域 $\mathbf{D}_g \subset \mathbb{R}^n$ 上的 n 元 m 维向量值函数。如果 $\mathbf{g}(\mathbf{D}_g) \subset \mathbf{D}_f$ ，那么可以构造复合函数

$$z = f \circ \mathbf{g} = f[y_1(x_1, \cdots, x_n), \cdots, y_m(x_1, \cdots, x_n)], \quad (x_1, \cdots, x_n) \in \mathbf{D}_g.$$

链式规则（矩阵表达）

链式规则 设 $\mathbf{g}$ 在 $\mathbf{x}^0 \in \mathbf{D}_g$ 点可导, 即 $y_1, \dots, y_m$ 在 $\mathbf{x}^0$ 点可偏导, 且 f 在 $\mathbf{y}^0 = \mathbf{g}(\mathbf{x}^0)$ 点可微, 则

$$\frac{\partial z}{\partial x_i}(\mathbf{x}^0) = \frac{\partial z}{\partial y_1}(\mathbf{y}^0) \frac{\partial y_1}{\partial x_i}(\mathbf{x}^0) + \dots + \frac{\partial z}{\partial y_m}(\mathbf{y}^0) \frac{\partial y_m}{\partial x_i}(\mathbf{x}^0), \quad i = 1, 2, \dots, n.$$

链式规则（矩阵表达）

上式可以用矩阵表示为

$$\left(\frac{\partial z}{\partial x_1}, \dots, \frac{\partial z}{\partial x_n} \right)_{x=\mathbf{x}^0} = \left(\frac{\partial z}{\partial y_1}, \dots, \frac{\partial z}{\partial y_m} \right)_{y=\mathbf{y}^0} \begin{pmatrix} \frac{\partial y_1}{\partial x_1} & \dots & \frac{\partial y_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial y_m}{\partial x_1} & \dots & \frac{\partial y_m}{\partial x_n} \end{pmatrix}_{x=\mathbf{x}^0}$$

或用向量值函数的导数记号表示为： $(f \circ \mathbf{g})'(\mathbf{x}_0) = f'(\mathbf{y}^0)\mathbf{g}'(\mathbf{x}^0).$

例 1.2

例 1.2 设 $z = \arctan(xy)$, $y = e^x$, 求 $\left. \frac{dz}{dx} \right|_{x=0}$ 。

例 1.2

解 由链式规则:

$$\frac{dz}{dx} = \frac{\partial z}{\partial x} \cdot \frac{dx}{dx} + \frac{\partial z}{\partial y} \cdot \frac{dy}{dx} = \frac{y}{1+x^2y^2} \cdot 1 + \frac{x}{1+x^2y^2} \cdot e^x = \frac{e^x(1+x)}{1+x^2e^{2x}}$$

于是

$$\left. \frac{dz}{dx} \right|_{x=0} = 1.$$

例 1.3

例 1.3 设 $z = \frac{x^2}{y}$, 而 $x = u - 2v$, $y = 2u + v$, 计算 $\frac{\partial z}{\partial u}$, $\frac{\partial z}{\partial v}$ 。

例 1.3

解

$$\begin{aligned}\frac{\partial z}{\partial u} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial u} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial u} = \frac{2x}{y} \cdot 1 + \left(-\frac{x^2}{y^2}\right) \cdot 2 \\ &= \frac{2(u-2v)}{2u+v} - \frac{2(u-2v)^2}{(2u+v)^2} = \frac{2(u-2v)(u+3v)}{(2u+v)^2}.\end{aligned}$$

$$\begin{aligned}\frac{\partial z}{\partial v} &= \frac{\partial z}{\partial x} \cdot \frac{\partial x}{\partial v} + \frac{\partial z}{\partial y} \cdot \frac{\partial y}{\partial v} = \frac{2x}{y} \cdot (-2) + \left(-\frac{x^2}{y^2}\right) \cdot 1 \\ &= -\frac{4(u-2v)}{2u+v} - \frac{(u-2v)^2}{(2u+v)^2} = \frac{(2v-u)(9u+2v)}{(2u+v)^2}.\end{aligned}$$

例 1.4

例 1.4 设 $z = (2x + y)^{x+2y}$, 计算 $\frac{\partial z}{\partial x}$, $\frac{\partial z}{\partial y}$ 。

例 1.4

解 $z = (2x + y)^{x+2y}$, 设 $u = 2x + y$, $v = x + 2y$, 则 $z = u^v$ 。于是

$$\begin{aligned}\frac{\partial z}{\partial x} &= \frac{\partial z}{\partial u} \frac{\partial u}{\partial x} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial x} = vu^{v-1} \cdot 2 + u^v \ln u \cdot 1 \\ &= 2(x + 2y)(2x + y)^{x+2y-1} + (2x + y)^{x+2y} \ln(2x + y) \\ &= (2x + y)^{x+2y} \left(\frac{2(x + 2y)}{2x + y} + \ln(2x + y) \right)\end{aligned}$$

例 1.4

$$\begin{aligned}\frac{\partial z}{\partial y} &= \frac{\partial z}{\partial u} \frac{\partial u}{\partial y} + \frac{\partial z}{\partial v} \frac{\partial v}{\partial y} = vu^{v-1} \cdot 1 + u^v \ln u \cdot 2 \\ &= (x+2y)(2x+y)^{x+2y-1} + 2(2x+y)^{x+2y} \ln(2x+y) \\ &= (2x+y)^{x+2y} \left(\frac{x+2y}{2x+y} + 2 \ln(2x+y) \right)\end{aligned}$$

例 1.5

例 1.5 设 $w = f(x^2 + y^2 + z^2, xyz)$, f 有二阶连续偏导, 求

$$\frac{\partial w}{\partial x}, \quad \frac{\partial^2 w}{\partial x \partial z}$$

例 1.5

解 记 $u = x^2 + y^2 + z^2$, $v = xyz$, 则 $w = f(u, v)$ 。显然

$$\frac{\partial u}{\partial x} = 2x, \quad \frac{\partial v}{\partial x} = yz$$

由链式规则

$$\frac{\partial w}{\partial x} = \frac{\partial w}{\partial u} \cdot \frac{\partial u}{\partial x} + \frac{\partial w}{\partial v} \cdot \frac{\partial v}{\partial x} = 2x \frac{\partial w}{\partial u} + yz \frac{\partial w}{\partial v}$$

例 1.5

注意 $\frac{\partial w}{\partial u}, \frac{\partial w}{\partial v}$ 仍为复合函数, 且: $\frac{\partial u}{\partial z} = 2z, \quad \frac{\partial v}{\partial z} = xy$ 。再用链式法则, 有:

$$\begin{aligned}\frac{\partial^2 w}{\partial x \partial z} &= \frac{\partial}{\partial z} \left(\frac{\partial w}{\partial x} \right) = \frac{\partial}{\partial z} \left(2x \frac{\partial w}{\partial u} + yz \frac{\partial w}{\partial v} \right) \\&= 2x \frac{\partial}{\partial z} \left(\frac{\partial w}{\partial u} \right) + y \frac{\partial w}{\partial v} + yz \frac{\partial}{\partial z} \left(\frac{\partial w}{\partial v} \right) \\&= 2x \left(2z \frac{\partial^2 w}{\partial u^2} + xy \frac{\partial^2 w}{\partial u \partial v} \right) + y \frac{\partial w}{\partial v} + yz \left(2z \frac{\partial^2 w}{\partial v \partial u} + xy \frac{\partial^2 w}{\partial v^2} \right) \\&= 4xz \frac{\partial^2 w}{\partial u^2} + 2y(x^2 + z^2) \frac{\partial^2 w}{\partial v \partial u} + xy^2 z \frac{\partial^2 w}{\partial v^2} + y \frac{\partial w}{\partial v}\end{aligned}$$

例 1.5

若用函数符号加下标 i 表示对其第 i 个变量的偏导数，即

$$f_1 = \frac{\partial f(u, v)}{\partial u}, f_2 = \frac{\partial f(u, v)}{\partial v}, f_{11} = \frac{\partial^2 f(u, v)}{\partial u^2}, f_{12} = \frac{\partial^2 f(u, v)}{\partial u \partial v}, f_{22} = \frac{\partial^2 f(u, v)}{\partial v^2}$$

如此等等，那么上面的结果可以表示为

$$\frac{\partial w}{\partial x} = 2xf_1 + yzf_2$$

$$\frac{\partial^2 w}{\partial x \partial z} = 4xz f_{11} + 2y(x^2 + z^2) f_{12} + xy^2 z f_{22} + y f_2$$

例 1.6

例 1.6 已知 $u = u(x, y)$ 为可微函数，求

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2$$

在极坐标下的表达式。

例 1.6

解 直角坐标和极坐标有如下关系: $x = r \cos \theta$, $y = r \sin \theta$, 将 x, y 看作中间变量则:

$$\frac{\partial u}{\partial r} = \frac{\partial u}{\partial x} \cos \theta + \frac{\partial u}{\partial y} \sin \theta \quad (1)$$

$$\frac{\partial u}{\partial \theta} = -r \sin \theta \frac{\partial u}{\partial x} + r \cos \theta \frac{\partial u}{\partial y} \quad (2)$$

例 1.6

将第一式乘 r 后的平方加上第二式的平方乘以 $1/r^2$ 得，既 $(1)^2 + \frac{1}{r^2}(2)^2$ ：

$$\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial u}{\partial y}\right)^2 = \left(\frac{\partial u}{\partial r}\right)^2 + \frac{1}{r^2} \left(\frac{\partial u}{\partial \theta}\right)^2$$

复合向量值函数的导数 *

设

$$\mathbf{f}: \mathbf{D}_f \rightarrow \mathbb{R}^2,$$

$$(u, v) \mapsto (x(u, v), y(u, v)),$$

是区域 $\mathbf{D}_f \subset \mathbb{R}^2$ 上的二元二维向量值函数。又设

$$\mathbf{g}: \mathbf{D}_g \rightarrow \mathbb{R}^2,$$

$$(s, t) \mapsto (u(s, t), v(s, t))$$

是区域 $\mathbf{D}_g \subset \mathbb{R}^2$ 上的二元二维向量值函数。

复合向量值函数的导数

如果 $\mathbf{g}$ 的值域 $\mathbf{g}(\mathbf{D}_{\mathbf{g}}) \subset \mathbf{D}_{\mathbf{f}}$, 则可以构造复合向量值函数 $\mathbf{f} \circ \mathbf{g}$ 。具体写出来就是

$$\begin{cases} x = x[u(s, t), v(s, t)], \\ y = y[u(s, t), v(s, t)], \end{cases} \quad (s, t) \in \mathbf{D}_{\mathbf{g}}.$$

复合向量值函数的导数 (续)

如果 $\mathbf{f}$ 和 $\mathbf{g}$ 分别在 $\mathbf{D}_f$ 与 $\mathbf{D}_g$ 上具有连续导数, 那么由链式法则

$$\frac{\partial x}{\partial s}(s, t) = \frac{\partial x}{\partial u}(u, v) \frac{\partial u}{\partial s}(s, t) + \frac{\partial x}{\partial v}(u, v) \frac{\partial v}{\partial s}(s, t),$$

$$\frac{\partial x}{\partial t}(s, t) = \frac{\partial x}{\partial u}(u, v) \frac{\partial u}{\partial t}(s, t) + \frac{\partial x}{\partial v}(u, v) \frac{\partial v}{\partial t}(s, t),$$

$$\frac{\partial y}{\partial s}(s, t) = \frac{\partial y}{\partial u}(u, v) \frac{\partial u}{\partial s}(s, t) + \frac{\partial y}{\partial v}(u, v) \frac{\partial v}{\partial s}(s, t),$$

$$\frac{\partial y}{\partial t}(s, t) = \frac{\partial y}{\partial u}(u, v) \frac{\partial u}{\partial t}(s, t) + \frac{\partial y}{\partial v}(u, v) \frac{\partial v}{\partial t}(s, t).$$

复合向量值函数的导数 (续)

写成矩阵形式就是

$$\begin{pmatrix} \frac{\partial x}{\partial s}(s, t) & \frac{\partial x}{\partial t}(s, t) \\ \frac{\partial y}{\partial s}(s, t) & \frac{\partial y}{\partial t}(s, t) \end{pmatrix} = \begin{pmatrix} \frac{\partial x}{\partial u}(u, v) & \frac{\partial x}{\partial v}(u, v) \\ \frac{\partial y}{\partial u}(u, v) & \frac{\partial y}{\partial v}(u, v) \end{pmatrix} \begin{pmatrix} \frac{\partial u}{\partial s}(s, t) & \frac{\partial u}{\partial t}(s, t) \\ \frac{\partial v}{\partial s}(s, t) & \frac{\partial v}{\partial t}(s, t) \end{pmatrix},$$

即

$$(\mathbf{f} \circ \mathbf{g})'(s, t) = \mathbf{f}'(u, v) \mathbf{g}'(s, t).$$

链式法则 (雅各比矩阵)

雅各比矩阵定义

设 $\mathbf{f}: \mathbf{D}_f(\subset \mathbb{R}^k) \rightarrow \mathbb{R}^m$ 与 $\mathbf{g}: \mathbf{D}_g(\subset \mathbb{R}^n) \rightarrow \mathbb{R}^k$ 分别是多元向量值函数, 且分别在 $\mathbf{D}_f$ 和 $\mathbf{D}_g$ 上具有连续导数。如果 $\mathbf{g}$ 的值域 $\mathbf{g}(\mathbf{D}_g) \subset \mathbf{D}_f$, 并记为 $u = \mathbf{g}(x)$, 那么复合函数 $\mathbf{f} \circ \mathbf{g}$ 在 $\mathbf{D}_g$ 上也具有连续的导数, 并且成立等式

$$(\mathbf{f} \circ \mathbf{g})'(x) = \mathbf{f}'(u)\mathbf{g}'(x) = \mathbf{f}'[\mathbf{g}(x)] \cdot \mathbf{g}'(x)$$

其中 $\mathbf{f}'(u)$, $\mathbf{g}'(x)$ 和 $(\mathbf{f} \circ \mathbf{g})'(x)$ 是相应的导数, 即雅各比矩阵。

例 1.7

设向量值函数 $\mathbf{f}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, 其坐标分量函数为

$$\begin{cases} x = \cos u \sin v, \\ y = \sin u \cos v \end{cases}$$

向量值函数 $\mathbf{g}: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, 其坐标分量函数为

$$\begin{cases} u = s + t, \\ v = s - t \end{cases}$$

例 1.7

则复合函数 $\mathbf{f} \circ \mathbf{g}$ 的坐标分量函数为

$$\begin{cases} x = \cos(s+t) \sin(s-t), \\ y = \sin(s+t) \cos(s-t) \end{cases}$$

例 1.7

解 构造偏导矩阵:

$$\begin{aligned}\begin{pmatrix} \frac{\partial x}{\partial s} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial s} & \frac{\partial y}{\partial t} \end{pmatrix} &= \begin{pmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{pmatrix} \begin{pmatrix} \frac{\partial u}{\partial s} & \frac{\partial u}{\partial t} \\ \frac{\partial v}{\partial s} & \frac{\partial v}{\partial t} \end{pmatrix} \\ &= \begin{pmatrix} -\sin u \sin v & \cos u \cos v \\ \cos u \cos v & -\sin u \sin v \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \\ &= \begin{pmatrix} -\sin u \sin v + \cos u \cos v & -\sin u \sin v - \cos u \cos v \\ \cos u \cos v - \sin u \sin v & \cos u \cos v + \sin u \sin v \end{pmatrix} \\ &= \begin{pmatrix} \cos(u+v) & -\cos(u-v) \\ \cos(u+v) & \cos(u-v) \end{pmatrix} = \begin{pmatrix} \cos 2s & -\cos 2t \\ \cos 2s & \cos 2t \end{pmatrix}.\end{aligned}$$

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一阶全微分的形式不变性

一阶全微分的形式不变性

以下总假设讨论的函数满足相应的可微条件。

设 $z = f(x, y)$ 为二元函数，那么当 x, y 为自变量时，

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy.$$

而当 x, y 为中间变量时，如

$$x = x(u, v), \quad y = y(u, v),$$

这时 $dx = x_u du + x_v dv$, $dy = y_u du + y_v dv$,

一阶全微分的形式不变性 (续)

一阶全微分的形式不变性 (续)

那么由链式法则得

$$\begin{aligned} dz &= \frac{\partial z}{\partial u} du + \frac{\partial z}{\partial v} dv \\ &= (f_x x_u + f_y y_u) du + (f_x x_v + f_y y_v) dv \\ &= f_x (x_u du + x_v dv) + f_y (y_u du + y_v dv) \\ &= \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy. \end{aligned}$$

这说明了无论 x, y 是自变量, 还是中间变量, 一阶微分具有相同的形式, 这就是一阶全微分的形式不变性。

多元函数的一阶全微分

多元函数情形

对于多元函数 $z = f(\mathbf{y})$, 其中 $\mathbf{y} = (y_1, y_2, \dots, y_m)^T$ 。当 $\mathbf{y}$ 为自变量时, 一阶全微分形式为

$$dz = f'(\mathbf{y})d\mathbf{y}.$$

而当 $\mathbf{y}$ 为中间变量 $\mathbf{y} = \mathbf{g}(\mathbf{x})$ ($\mathbf{x} = (x_1, x_2, \dots, x_n)^T$) 时, $d\mathbf{y} = \mathbf{g}'(\mathbf{x})d\mathbf{x}$ 。由链式法则,

$$dz = (f \circ \mathbf{g})'(\mathbf{x})d\mathbf{x} = f'(\mathbf{y})\mathbf{g}'(\mathbf{x})d\mathbf{x} = f'(\mathbf{y})(\mathbf{g}'(\mathbf{x})d\mathbf{x}) = f'(\mathbf{y})d\mathbf{y}.$$

这说明一阶全微分的形式不变性是普遍成立的。

例 2.1

例 2.1 计算函数 $u = x + \sin \frac{y}{2} + e^{yz}$ 的全微分。

计算全微分 例 2.1(解)

解 1

$$\frac{\partial u}{\partial x} = 1, \quad \frac{\partial u}{\partial y} = \frac{1}{2} \cos \frac{y}{2} + ze^{yz}, \quad \frac{\partial u}{\partial z} = ye^{yz},$$

所求全微分

$$du = dx + \left(\frac{1}{2} \cos \frac{y}{2} + ze^{yz} \right) dy + ye^{yz} dz.$$

解 2 直接求微分

$$\begin{aligned} du &= dx + d\left(\sin \frac{y}{2}\right) + d(e^{yz}) \\ &= dx + \frac{1}{2} \cos \frac{y}{2} dy + ze^{yz} dy + ye^{yz} dz \end{aligned}$$

例 2.2

例 2.2 求函数 $z = (x - y)e^{xy}$ 的偏导数和全微分。

例 2.2(解)

解

$$\begin{aligned} dz &= d[(x - y)e^{xy}] = (x - y)de^{xy} + e^{xy}d(x - y) \\ &= (x - y)e^{xy}(ydx + xdy) + e^{xy}(dx - dy) \\ &= e^{xy}(xy - y^2 + 1)dx + e^{xy}(x^2 - xy - 1)dy, \end{aligned}$$

所以

$$\frac{\partial z}{\partial x} = e^{xy}(xy - y^2 + 1), \quad \frac{\partial z}{\partial y} = e^{xy}(x^2 - xy - 1).$$

例 2.3

例 2.3 求函数 $z = x \ln(x^2 - 2y)$ 的偏导数和全微分。

例 2.3(解)

解

$$\begin{aligned} dz &= d[x \ln(x^2 - 2y)] \\ &= \ln(x^2 - 2y)dx + x d[\ln(x^2 - 2y)] \\ &= \ln(x^2 - 2y)dx + x \frac{d(x^2 - 2y)}{x^2 - 2y} \\ &= \left[\ln(x^2 - 2y) + \frac{2x^2}{x^2 - 2y} \right] dx - \frac{2x}{x^2 - 2y} dy, \end{aligned}$$

所以

$$\frac{\partial z}{\partial x} = \ln(x^2 - 2y) + \frac{2x^2}{x^2 - 2y}, \quad \frac{\partial z}{\partial y} = -\frac{2x}{x^2 - 2y}.$$

例 2.4

例 2.4 设 $z = \sin u \cos v$ 、 $u = xy$ 、 $v = \frac{y}{x}$ ，求 $\frac{\partial z}{\partial x}$ 及 $\frac{\partial z}{\partial y}$ 。

例 2.4(解)

解

$$\begin{aligned} dz &= \cos u \cos v du - \sin u \sin v dv \\ &= \cos u \cos v (y dx + x dy) - \sin u \sin v \left(-\frac{y}{x^2} dx + \frac{1}{x} dy \right) \\ &= \left(y \cos u \cos v + \frac{y}{x^2} \sin u \sin v \right) dx + \left(x \cos u \cos v - \frac{1}{x} \sin u \sin v \right) dy \\ \frac{\partial z}{\partial x} &= y \cos u \cos v + \frac{y}{x^2} \sin u \sin v, \quad \frac{\partial z}{\partial y} = x \cos u \cos v - \frac{1}{x} \sin u \sin v \end{aligned}$$

例 2.5

例 2.5 设 $z = \sqrt[4]{\frac{x+y}{x-y}}$, 求全微分 dz 。

例 2.5(解)

解 在 $z = \sqrt[4]{\frac{x+y}{x-y}}$ 的两边取对数,

$$\ln z = \frac{1}{4}[\ln(x+y) - \ln(x-y)].$$

两边求全微分, 利用一阶全微分的形式不变性, 得到

$$\frac{dz}{z} = \frac{1}{4} \left[\frac{dx + dy}{x + y} - \frac{dx - dy}{x - y} \right],$$

例 2.5(解)

即

$$dz = \frac{1}{2} \sqrt[4]{\frac{x+y}{x-y}} \cdot \frac{x dy - y dx}{x^2 - y^2}.$$

$$\therefore \frac{\partial z}{\partial x} = -\frac{1}{2} \sqrt[4]{\frac{x+y}{x-y}} \cdot \frac{y}{x^2 - y^2}, \quad \frac{\partial z}{\partial y} = \frac{1}{2} \sqrt[4]{\frac{x+y}{x-y}} \cdot \frac{x}{x^2 - y^2}.$$

关于高阶微分的讨论

要注意的是，全微分的形式不变性在高阶微分时是不成立的。例如函数 $z = f(x, y)$ ，当 x, y 为自变量时，

$$d^2z = \frac{\partial^2 z}{\partial x^2} dx^2 + 2 \frac{\partial^2 z}{\partial x \partial y} dx dy + \frac{\partial^2 z}{\partial y^2} dy^2,$$

而当 x, y 为中间变量时

$$d^2z = \frac{\partial^2 z}{\partial x^2} dx^2 + 2 \frac{\partial^2 z}{\partial x \partial y} dx dy + \frac{\partial^2 z}{\partial y^2} dy^2 + \frac{\partial z}{\partial x} d^2x + \frac{\partial z}{\partial y} d^2y.$$

与一维情况相同，当 x, y 为自变量时， $d^2x = d^2y = 0$ ；而当 x, y 为中间变量时， d^2x 与 d^2y 一般不为零。

例 2.6

例 2.6 设 $z = \ln(x + y)$, 求 $d^k z$ 。

例 2.6(解)

解 设 $u = x + y$, 则 $du = dx + dy$, $d^2u = 0$, 因此 $d^k u = 0, k \geq 2$ 。于是

$$dz = \frac{du}{u} = \frac{dx + dy}{x + y},$$

$$d^2z = d\left(\frac{du}{u}\right) = d\left(\frac{1}{u}\right) du + \frac{1}{u} d^2u = -\frac{du^2}{u^2} = -\frac{(dx + dy)^2}{(x + y)^2},$$

$$d^3z = d\left(-\frac{du^2}{u^2}\right) = d\left(-\frac{1}{u^2}\right) du^2 - \frac{1}{u^2} d(du^2) = \frac{2du^3}{u^3} = \frac{2!(dx + dy)^3}{(x + y)^3}.$$

应用归纳法即可得到

$$d^k z = (-1)^{k+1} \frac{(k-1)!(dx + dy)^k}{(x + y)^k}, \quad k = 1, 2, \dots$$

目录

- ▶ 多元复合函数的求导法则
- ▶ 复合函数的全微分
- ▶ 隐函数的偏导数

第十七章 隐函数存在定理

作业

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